

Global Outlook

All energy types will be needed

- Oil is projected to remain the largest source of primary energy, as it is essential for industrial manufacturing, including as a raw material, and commercial transportation.
- Natural gas demand is projected to rise, largely to help meet increasing needs for electricity and lower-emission industrial heat.
- Lower-emission energy sources, including solar, wind, and biofuels, are expected to have the fastest rate of increase.
- By 2050, the world is expected to use 35% less coal than it does today as lower-emission sources meet an increasing share of rising demand in the developing world.

Report

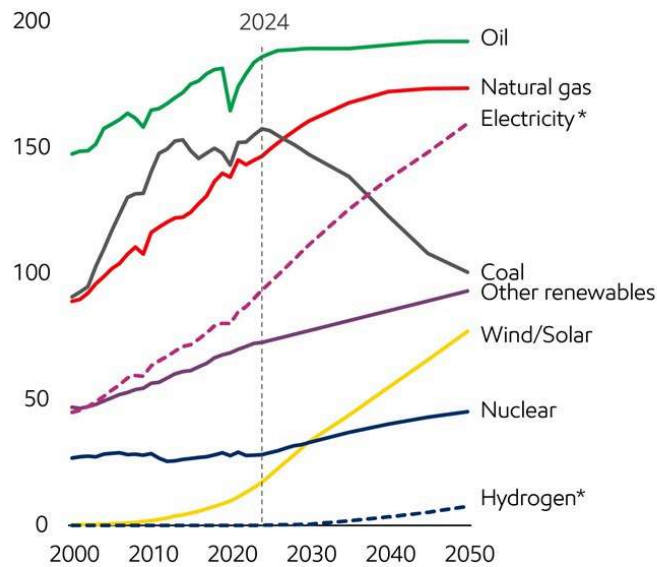
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The Global Outlook projects that oil and natural gas will be the largest energy sources in 2050, and make up more than half of the world's energy supply. The

biggest change in the world's energy mix between now and 2050 will be a significant increase in solar and wind, coupled with a large reduction in coal.

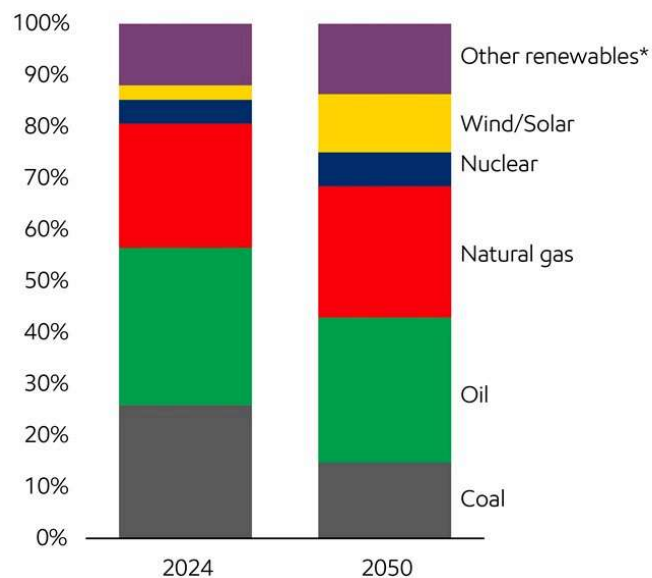
Global energy demand by fuel

Primary energy - Quadrillion Btu



*Electricity and Hydrogen are secondary energies derived from the primary energies shown

Percent of primary energy



*includes biomass, biofuels, hydropower, geothermal

Demand for oil is projected to grow to ~105 MBD by 2050, up from ~100 MBD today, while demand for natural gas is projected to grow to ~500 BCFD in 2050, an increase of 20% compared to current levels. The Outlook's projection for oil and gas demand is driven by the **unique needs of industry and commercial**

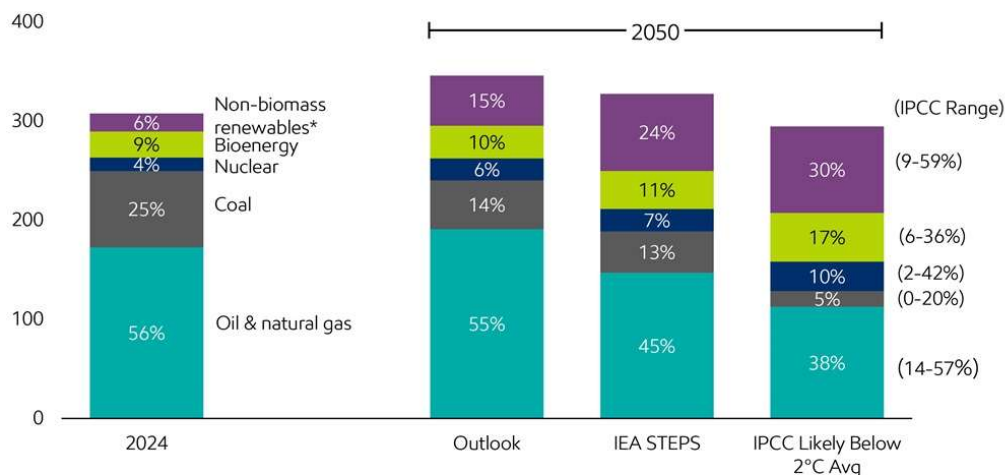
transportation which, combined, make up 75% of global oil demand, >40% of natural gas demand (excluding off-site electricity), and ~50% of electricity demand in 2050.

Global electricity demand is expected to grow 70% by 2050, driven by improving living standards in developing countries. Renewables see significant growth with solar and wind generation projected to increase more than fourfold, from <15% of the world's electricity generation to >40%. Coal use will shrink as it is displaced by lower-emission energy sources – not just renewables but also natural gas, which has about half the carbon intensity of coal in electricity generation (EIA, Natural Gas Explained).

How does ExxonMobil's Global Outlook compare to other projections or scenarios?

Global energy mix

Million barrels per day of oil equivalent



*Non-biomass renewables includes hydro, wind, solar, and geothermal

Source: 2024 IEA World Energy Outlook; IPCC: AR6 Scenarios Database hosted by IIASA release 1.0 average IPCC C3: "Likely below 2°C" scenarios

The IPCC's average of Likely Below 2°C scenarios include considerably larger efficiency gains, which lower overall energy demand, and faster deployment of renewables, which grow to ~30% of total global energy. Under these scenarios, oil and natural gas still play the largest role in 2050 at 38% of total global energy supply.

Averages can mask large underlying differences, so it is also important to look at the *range* of energy-mix outcomes across the IPCC's 311 Likely Below 2°C scenarios. Some include coal being completely eliminated by 2050. Some include a revival of nuclear that brings this emissions-free energy source to 42% of the world's total energy. Some include renewables growing approximately 10x to become the world's dominant energy source. And some include oil and natural gas

remaining the largest energy source and actually delivering *more* of the world's energy supply than they do today, while still hitting the likely below 2°C target.

Sensitivity: Road transportation fuel demand

As **our look-back showed**, there is uncertainty in all projections of future energy demand. Let's explore how electrification of road transportation could impact projections of oil demand.

Passenger cars

Sales of electric vehicles have grown significantly in recent years, now making up ~20% of total sales. However, the mix of electric vehicles has also shifted, with plug-in hybrids now making up more than a third of electric vehicle sales.

*Includes gasoline, diesel, and CNG/LPG vehicles

Despite rapidly growing sales, the global passenger car fleet is still only <5% electric vehicles (including BEV, PHEV, and fuel cell), as it takes ~15 years for the fleet to turnover. Our Outlook projects that by 2050, electric cars will make up ~50% of global sales, and ~40% of the global passenger car fleet.

Oil excl. biofuels demand back to '10 levels in sensitivity

Million barrels per day

This reflects an ExxonMobil hypothetical sensitivity; not a base case

Energy-related CO₂ emissions down 5% in sensitivity

Billion tonnes

This reflects an ExxonMobil hypothetical sensitivity; not a base case

Passenger car sensitivity analysis: But what would be the impact on oil demand if EV sales grow faster than our Outlook projects?

The Global Outlook projects battery-electric vehicles to be ~25% of all new car sales by 2035 and ~30% by 2050 (excluding plug-in hybrid electric vehicles and fuel cell). Compare that to this sensitivity, which assumes 100% battery electric vehicle

sales from 2035 onward, resulting in an almost fully electrified global car fleet by 2050.

This 100% electric fleet would reduce global demand for oil back to the same levels they were in 2010. Global CO₂ emissions decrease ~5% versus the Global Outlook, with the decline in light-duty CO₂ emissions partially offset by emissions from increased power generation.

Heavy-duty trucks

Heavy-duty truck sensitivity analysis: What if technology advances enabled faster transition to lower-emission freight solutions?

This technology sensitivity evaluates sales reaching nearly 100% electrification of light commercial vehicles, about 70% alternative fuels in medium commercial vehicles, and about 20% penetration of alternative fuels in long-haul commercial vehicles by 2050.

This technology sensitivity results in 2050 liquids demand similar to current levels, as the growth in electrification, the use of hydrogen, and efficiency gains offset the growing demand for commercial transportation.

Heavy-duty fuel demand

Million barrels per day of oil equivalent

Alternate EV sales assumption: 100% LCV by 2035, and 70% MCV and 20% HCV by 2040

This reflects an ExxonMobil hypothetical sensitivity, not a base case

Fuel share analysis is based on IEA Transport Project, ExxonMobil analysis

Oil (excl. bio) demand with heavy-duty truck sensitivity

Million barrels per day

Technology development to drive continued efficiency gains is also important. An alternate (upside) sensitivity explores the impact if no further fuel efficiency gains are made (vs. our Outlook which projects continued improvement at historical

trends of ~0.5% per year). This sensitivity results in an ~10% increase of heavy-duty fuel demand in 2050.

Affordability will drive the pace of any energy transition

Sustained economic growth and continued innovation to reduce costs for key technologies are essential for improving affordability.



Sustained oil and gas investment is more important than ever

Oil and natural gas supply from producing wells naturally declines over time, which requires investment in new and existing fields to meet demand across scenarios.



Global Outlook

Learn more about our view to 2050 →